

NMR Probe with Short Recovery Time

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A new NMR probe is described for use at low frequencies with an active damping circuit, which uses a quarter wave network and diode expander to dampen the ringing of the probe after RF excitation. The damping does not operate during the reception of the nuclear signal and therefore there is no loss in sensitivity as would be the case for permanent damping. The high voltage portion of the ringdown is damped by means of a resistor. The probe is most efficient at low frequencies and a fivefold reduction in recovery time to 10 μ s has been obtained at 7 MHz ($Q = 58$). At this low frequency, the signal/noise ratio has been increased by using a sample coil with two segments connected in parallel. The damper pulse generator is also described. © 1987 Academic Press, Inc.

INTRODUCTION

This paper describes a relatively simple probe with an active damping circuit which is useful for NMR studies of solids at low frequencies, for example, when studying nuclei of low gyromagnetic ratio.

The probe of an ideal NMR pulse spectrometer should possess the following features: (1) For the duration of the strong RF pulse the impedance of the probe should be adjusted to the output impedance of the transmitter in order to ensure the maximum power transfer. The receiving system should be isolated from the probe to protect it from large RF voltages. (2) During reception of the weak nuclear signal, the probe should be isolated from the transmitter output and efficiently coupled to the receiving system. In this way the receiver is protected from any influence of transmitter noise and the signal is driven to the receiving system only. (3) The probe should have a short recovery time. (4) The signal-to-noise ratio (S/N) of the nuclear signal should be as high as possible.

Feature (3) is in conflict with (4) because the natural ringing time constant τ depends on the quality factor Q

$$\tau = 2Q/\omega_0$$

and needs to be as short as possible, while $S/N \propto \sqrt{Q}$. Typically one must wait a

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recovery time $T_R \simeq 20 \tau$ for the circuit to ring down to noise level following a 1 kV RF pulse at 10 k Ω . As an example for $Q = 58$, $\tau = 2.6 \mu\text{s}$ and $T_R \simeq 52 \mu\text{s}$ at 7 MHz. Many techniques for designing probes for use in pulsed-NMR studies of solids are described in the literature.

Techniques which use crossed coils (1-5) are difficult in design when we want to make experiments at low temperatures using a cryostat. Bridge circuits (4, 6-10) have some disadvantages because of their critical tuning conditions. Loose coupling schemes (11-14) do not have good isolation features. High-frequency techniques (15-17) are characterized by extremely fast recoveries but are not suitable for work at low frequencies. Many probe dampers employed to dissipate the RF energy stored in the coil have been described in the literature (2, 18-29) but only small improvements in the value of T_R are actually observed. The recovery time also depends on the receiver itself (21, 29-32) and on the transmitter itself (5, 27, 33). The scheme proposed by Chingas (34) for reduction of recovery time uses overcoupling techniques, which needs a transmitter which is unconditionally stable. Sullivan and collaborators (35) use an elegant technique which gave the very short recovery time of 7 μs at 3 MHz without loss of sensitivity. The scheme proposed by Hoult (27, 36) gave 7 μs at 5 MHz. These techniques by Sullivan *et al.* and Hoult are rather complicated.

The scheme proposed here is not as efficient as that of Sullivan *et al.*, but it is relatively simple and easy to adapt in a conventional NMR spectrometer. Moreover, the scheme does not introduce extra noise as in that proposed by Kisman and Armstrong (22). Our scheme is described in the next section, and the operation of the circuit and the damper pulse generator are described in the subsequent sections.

PROBE

Assume that the fall-time of the transmitter RF pulse in purely resistive load is shorter than the ringing time constant of the probe. If the recovery time of the receiving system is shorter than the time required for the RF pulse of peak amplitude V_0 across the sample coil to drop to a value comparable with the induced nuclear signal V_n , the recovery time of the probe is given by (15)

$$T_R = \frac{2Q}{\omega} \ln \frac{V_0}{V_n}.$$

As mentioned in the Introduction, decreasing the Q factor in the usual single coil causes a decrease of the S/N ratio of the incoming nuclear signal, but reduces the recovery time. Therefore, the use of single sample coils was excluded and instead a coil with two segments connected in parallel was used (37, 38). At fixed frequency such a coil gives a larger sample volume, decreased Q factor, and increased S/N ratio, which is very favorable for the reduction of recovery time at low frequencies. Moreover the coils have a relatively homogeneous B_1 field and the axial electric field at the sample can be reduced by connecting the coils in such a way that an electrostatic plane of mirror symmetry bisects the sample. Although originally intended for use at higher frequencies (37, 38), these coils also have good features for reduction in recovery time at low frequencies. It is worth adding that in comparison with a single coil, these coils have lower inductance L , which together with a lower Q factor causes the dy-

namical resistance of the coils $R_d = Q\omega L$ to be lower and thus the recovery time is also decreased in respect to the peak amplitude V_0 , since $V_0 \propto R_d$.

Quarter wavelength lines are generally used in conventional probes to protect the receiver input while the RF pulse is on. At low frequencies quarter wavelength transmission lines become inconveniently long (more than 8 m at 7 MHz) and thus the possibility of RF pickup during the receive time occurs. In our probe (Fig. 1) these quarter wavelength lines are replaced by quarter wave networks (39) made of capacitors and inductors which act like quarter wave lines for impedance matching. Such networks have advantages over $\lambda/4$ lines: the characteristic impedance (as well as the operating frequency) is adjustable, so that one such network can be adjusted to do the work of many cables; we eliminate long cables at lower frequencies giving better operation and the removing of an unmanageable mass of cables; during an RF pulse the voltage applied to the receiver input is typically only one-tenth of that obtained with standard $\lambda/4$ lines (40). Our probe (Fig. 1) contains two quarter wave networks. Quarter wave network 1 serves mainly to protect the receiver input, while quarter wave network 2, as seen presently, reduces the recovery time. The following equations describe the quarter wave network (39), with $L_1 = L_2 = L_3 = L_4 = L$ and $C_3 = C_4 = C$:

- (1) Characteristic impedance: $Z_c = \omega L$,
- (2) Relation between input impedance Z_i and output impedance Z_o : $Z_i Z_o = (\omega L)^2$,
- (3) Resonance condition: $\omega^2 LC = 1$.

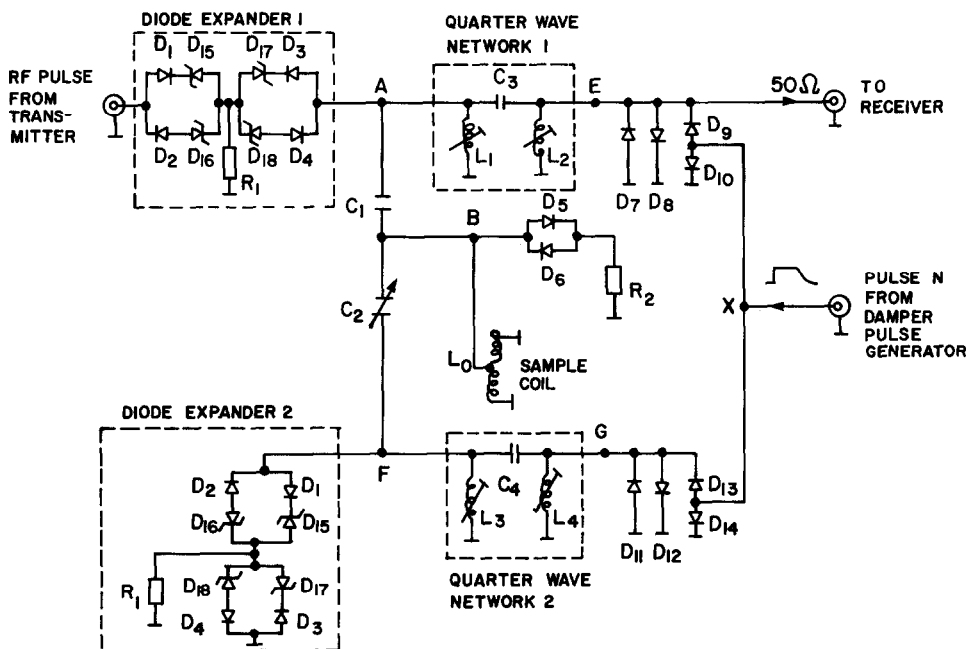


FIG. 1. Circuit diagram of the probe. Values of individual components have been chosen for 7 MHz. D_1 – D_{14} —1N914; D_{15} – D_{18} —6.8 volt Zener Diode Type UZ8806; $R_1 = 1\text{ K}\Omega$; $R_2 = 900\ \Omega$; $C_1 \cong 39\text{ pF}$; C_2 —variable capacitors required to resonate coil L_0 ; $C_3 = C_4 \cong 453\text{ pF}$; $L_0 = 1.35\ \mu\text{H}$; $L_1 = L_2 = L_3 = L_4 \cong 1.14\ \mu\text{H}$.

Our probe contains two diode expanders. The device has been so named because it serves the opposite function to a diode limiter (or clipper). Specifically, it blocks low-level signals (below the thresholds of the diodes) and passes high-level signals which are much larger than the threshold voltage of normal silicon diodes. The device is made from Zener diodes and ordinary diodes. A Zener diode acts like a regular diode which also conducts in the reverse direction when the voltage (in the reverse direction) exceeds a specified threshold which can be much higher than the 0.5 V forward threshold voltage for silicon diodes. The diode expander contains 6.8 V Zener diodes and therefore passes signals which are larger than 15 V either side of the zero crossing point. Diode expander 1 is placed between the transmitter and the probe input in order to prevent low level energy in the transmitter output from being coupled to the probe and receiver. Diode expander 2 serves to reduce recovery times as seen presently.

The resonant circuit of the probe (Fig. 1) consists of the inductance L_0 of the sample coil (two segments connected in parallel) and the capacitance C_1 and C_2 . The resonance condition for this circuit is

$$1/\omega(C_1 + C_2) = \omega L_0.$$

The value of capacitance C_1 is calculated for maximum sensitivity of the nuclear signal. The impedance $R' = 50 \Omega$ at point A is transformed to impedance R'' , which is placed across the sample coil at point B according to the relation (13, 36)

$$R'' = 1/R'(\omega C_1)^2. \quad [1]$$

For matching, R'' should be equal to $Q\omega L_0$, where Q is the quality factor of the coil. Setting

$$R'' = Q\omega L_0 \quad [2]$$

halves the net parallel resistance, and thus halves the Q of the circuit. Because the experimental value of Q for the unloaded coil is 116, during observation of the nuclear signal the value will be 58. The value of C_1 is given, from Eqs. [1] and [2], by

$$C_1 = 1/\omega(Q\omega L_0 R')^{1/2}. \quad [3]$$

Substituting in Eq. [3] the values $\omega = 2\pi \times 7 \times 10^6$ Hz, $Q = 116$, $L_0 = 1.35 \mu\text{H}$, and $R' = 50 \Omega$, we find that $C_1 = 38.7$ pF.

OPERATION OF THE CIRCUIT

Consider what happens during a transmitter pulse. For the duration of the strong RF pulse all the diodes of the probe conduct and have low impedances. Neglecting the voltage across the diode expanders, point A is connected to the transmitter output, point F is grounded, point E is grounded, resistor R_2 is connected to point B and thus the resonant circuit is damped giving a low $Q = 12$. The equivalent circuit of the probe when the RF is on is given in Fig. 2. Since point A is a quarter wavelength from E ($L_1 C_3$ connected in parallel resonance), the effective short circuit at E appears as an open circuit at A and thus does not affect the voltage at A and the output to the receiver is protected from the RF pulse. Simultaneously with the RF pulse, a pulse N from the damper pulse generator is applied to point X (Fig. 1) and therefore diodes

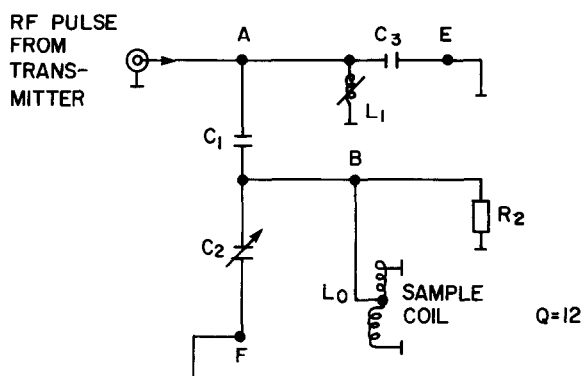


FIG. 2. Diagram of the equivalent circuit during RF pulse on. L_1C_3 connected in parallel.

D_9 , D_{10} , D_{13} , and D_{14} also conduct. The relationship between the RF pulse and other pulses and the nuclear signal are shown in Fig. 5. Pulse H shows the voltage applied to the transmitter modulator (pulse H produces an RF pulse) and the RF pulse voltage across the sample coil. Pulse N serves to reduce the recovery time. Trace Y shows changes of the quality factor Q . Next we focus attention on the recovery time. From t_0 to t_1 the pulse H produces an RF pulse which is applied to the probe. During this time the Q factor is decreased from 58 to 12 because diodes D_5 and D_6 are conducting and the circuit is damped by R_2 . From t_1 to t_2 the RF pulse rings down and the Q factor is still 12. When the amplitude of the RF pulse at B (Fig. 1) reaches about 15 V at t_2 (Fig. 5) and diodes D_9 , D_{10} , D_{13} , and D_{14} are still conducting by means of pulse N, then diode expander 2 becomes nonconducting (diode expander 1 becomes nonconducting a little earlier). In this situation the impedances at A and F in Fig. 1 become large and the quality factor Q is strongly decreased ($Q \ll 12$). Between t_2 and

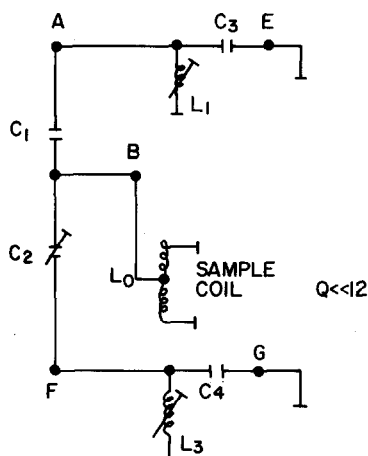


FIG. 3. Diagram of the equivalent circuit during damping. L_1C_3 and L_3C_4 connected in parallel.

t_3 (Fig. 5) when $Q \ll 12$, the amplitude of the RF pulse rings down very fast from 15 V to noise level. The equivalent circuit for this damping period t_2 to t_3 is shown in Fig. 3. Both quarter wave networks have an effective short circuit at points E and G, which appears as an open circuit at points A and F. In other words, L_1C_3 and L_3C_4 are connected in parallel presenting high impedances at points A and F. Between t_3 and t_4 (Fig. 5) the circuit is characterized by a gradually increasing quality factor Q because of the gradually decreasing current flowing through the diodes D_9 , D_{10} , D_{13} , and D_{14} (Fig. 1). From t_4 (Fig. 5) onward, Q remains high ($Q = 58$) and the free induction signal (FID) can be observed with good S/N ratio. The equivalent circuit during reception of the FID is shown in Fig. 4. Because the quarter wave network 1 has a characteristic impedance equal to 50Ω , impedance at points A and E are also 50Ω and are matched to the 50Ω input impedance of the receiver. Point F has a low impedance to ground because C_4L_4 are connected in series resonance. Optimization of the pulse N from the damper pulse generator (Fig. 6) in respect to its duration, slope, and amplitude gives the shortest recovery time. Experimentally it was found that for a 90° RF pulse of duration $t_1 - t_0 = 1.6 \mu\text{s}$ (Fig. 5), the recovery time $T_R = t_4 - t_1$ was equal to $10 \mu\text{s}$. It is clear from Fig. 5 that the system rings down to noise level within $3 \mu\text{s}$ but the rebuilding of the Q factor takes $7 \mu\text{s}$. Without damping the high voltage portion from t_1 to t_2 and without using the damping pulse N, the recovery time for $Q = 58$ would be approximately $53 \mu\text{s}$. Thus, the recovery time at 7 MHz has been reduced by a factor of 5. We have also tested this system at 14 MHz and 28 MHz, obtaining a reduction factor of recovery time of 4 and 1.8, respectively.

DAMPER PULSE GENERATOR

The damper pulse generator produces the pulse N which shortens the recovery time. This system is shown in Fig. 6. The potentiometer P_1 controls the pulse width at output K, potentiometer P_2 controls the slope of the trailing edge of the pulse, and

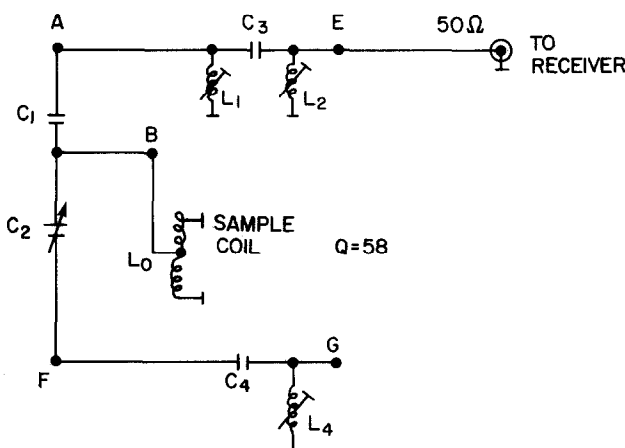


FIG. 4. Diagram of the equivalent circuit during receiving of the FID signal. Quarter wave network $L_1C_3L_2$ has characteristic impedance equal 50Ω . C_4L_4 connected in series.

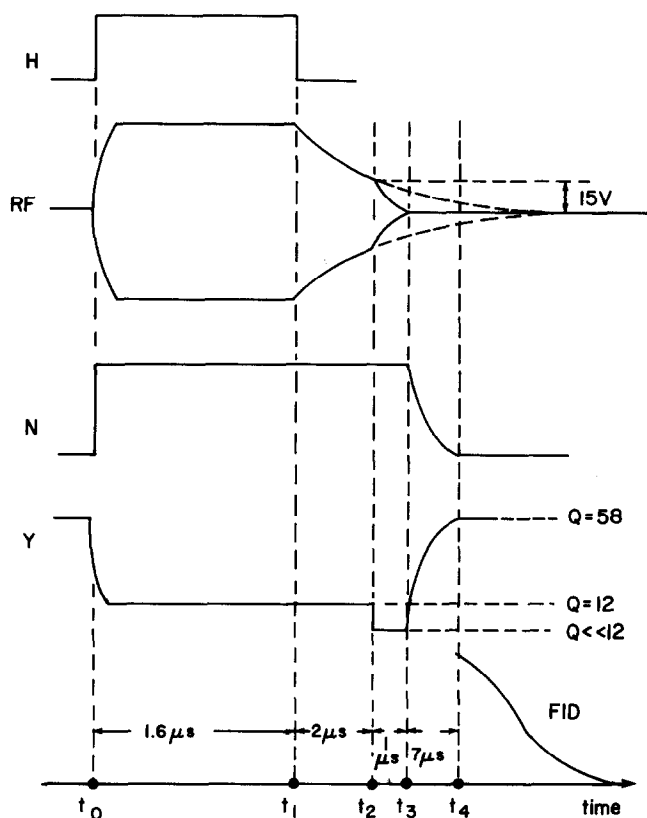


FIG. 5. Schematic representation of the recovery of the probe (not to scale).

potentiometer P_3 controls the amplitude of the pulse at output N. The timing of the pulses produced is shown in Fig. 7. The logic pulse at input H is synchronous with the RF pulse width.

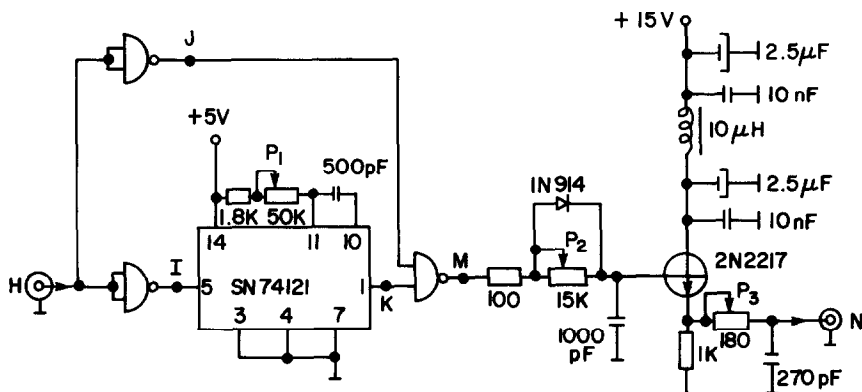


FIG. 6. Logic diagram of the damper pulse generator.

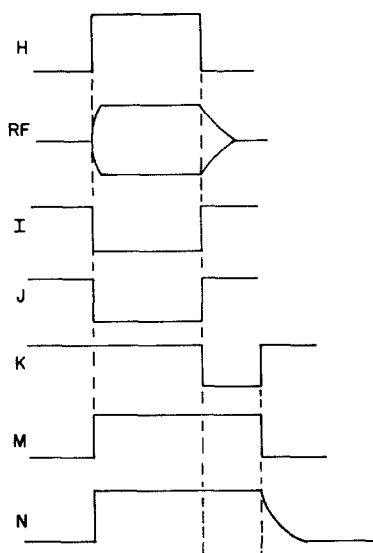


FIG. 7. Timing of the damper pulse generator.

FINAL REMARKS

We have described an NMR probe which shortens the recovery time at low frequencies. At 7 MHz the recovery time is reduced by a factor of 5 relative to the natural recovery time of the NMR circuit. The circuit for shortening the recovery time is simple and does not degrade the signal-to-noise ratio and can be readily adapted for a conventional NMR spectrometer probe. The NMR probe has good isolation features.

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